

LatentView Analytics

COMPREHENSIVE INTERVIEW QUESTION BANK (MULTI-ROLE)

This document compiles tailored interview questions for key data roles at LatentView Analytics. Questions balance core technical architecture, analytics problem-solving, and client-facing business frameworks specific to digital analytics consulting.

1. Data Analyst

Target Experience: 0–2 Years | Focus: SQL foundations, guesstimates, dashboarding basics, and communication

- **SQL Querying:** Write a robust SQL query to find the second-highest salary from an Employee table without using vendor-specific analytical functions like **LIMIT** or **TOP**.
- **Join Mechanics:** Explain a practical business scenario where a **LEFT JOIN** yields completely different analytical results than an **INNER JOIN**.
- **BI Architecture:** In Power BI, what is the core behavioral and performance difference between creating a Calculated Column versus defining a Measure?
- **Data Lookup:** Why is an **INDEX-MATCH** structure often preferred over **VLOOKUP** in large-scale Excel modeling?
- **Analytical Puzzle:** If the probability of an advertisement click occurring in a 20-minute window is 0.9, what is the probability of a click occurring within a compressed 5-minute window, assuming a uniform probability distribution?
- **Guesstimate:** Walk me through your framework to estimate the total number of commercial flights active in the sky over India at exactly 5:00 PM on a weekday.

2. Senior Data Analyst

Target Experience: 3–5+ Years | Focus: Query optimization, advanced BI frameworks, schema design, and root-cause analysis

- **Window Functions:** Distinguish between the behaviors of **ROW_NUMBER()**, **RANK()**, and **DENSE_RANK()** when handling duplicate row values.
- **Performance Tuning:** If a production SQL query joining 10-million-row transaction tables begins to experience severe latency, what systematic optimization steps would you execute?
- **Advanced Tableau:** Detail Level of Detail (LOD) expressions (**FIXED**, **INCLUDE**, **EXCLUDE**) and provide a business use case where standard dashboard filtering fails to achieve the correct aggregation.
- **Data Warehousing:** Compare a Star Schema against a Snowflake Schema. How does normalizing your dimensions into a Snowflake schema impact analytical query execution and dashboard load times?
- **Root-Cause Case Study:** A major e-commerce client experiences a sudden 15% drop in conversion rate over a weekend. Outline your step-by-step diagnostic framework to uncover the cause.

3. Data Scientist

Target Experience: 2–5 Years | Focus: Applied statistics, supervised learning, data preparation, and domain-specific modeling

- **Hypothesis Testing:** Clearly define Type I and Type II errors. How would you explain the real-world operational risk of a *p-value* threshold choice to a business stakeholder?
- **Algorithm Selection:** Under what operational constraints would you favor a traditional, interpretable Logistic Regression or Decision Tree over a high-performing non-linear ensemble like XGBoost?
- **Class Imbalance:** Detail your systematic preprocessing pipeline when building a classifier on a dataset where the positive target class represents only 0.5% of total records.
- **Marketing Mix Modeling (MMM):** Explain the concept of Market Mix Modeling. How do you mathematically model and capture the diminishing returns (saturation) and carryover effects (ad stock delay) of digital ad spend?

4. Senior Data Scientist

Target Experience: 5+ Years | Focus: MLOps, deep mathematical evaluation, feature engineering, and scalable ML architecture

- **Evaluation Metrics:** Mathematically differentiate between Log-Loss and ROC-AUC. Under what specific business objectives would you explicitly optimize for one over the other?
- **Multicollinearity:** How do you compute the Variance Inflation Factor (VIF)? If three critical features exhibit high multicollinearity, how do you remediate the issue without dropping predictive power?
- **MLOps Deployment:** Describe your end-to-end framework for deploying a real-time predictive model into a cloud environment. How do you architect automatic detection for data drift and concept drift?
- **System Design:** Design a highly scalable pipeline capable of ingesting high-throughput real-time streaming transactions and scoring them for fraudulent patterns with sub-100ms latency.

5. Data Engineer

Target Experience: 3+ Years | Focus: Big data pipelines, distributed systems, ETL/ELT optimization, and ingestion

- **Architecture Strategy:** When designing an enterprise-level analytics data lake, when do you choose an ELT pattern over a traditional ETL workflow?
- **Distributed Computing:** Explain how Apache Spark utilizes memory management. What is the fundamental difference between transformations and actions, and how does Spark build and optimize its Directed Acyclic Graph (DAG)?
- **Streaming Ingestion:** Architect a robust, fault-tolerant streaming ingestion pipeline utilizing Apache Kafka and Spark Streaming to handle high-velocity log files.
- **Data Storage:** Contrast columnar formats like Parquet/ORC against row-based formats like Avro. Which format is best optimized for analytical aggregation workloads, and why?

6. Data Science Specialist

Target Experience: Lead / Principal | Focus: Client advisory, structural ambiguity reduction, high-level system scoping

- **Ambiguity Resolution:** A Fortune 500 retail executive asks you to "leverage AI to maximize client lifetime value" but possesses siloed, unstructured historical data across four disconnected legacy systems. How do you define the MVP and timeline?
- **Advanced ML:** Contrast Gaussian Mixture Models (GMM) with K-Means clustering. Under what underlying topological distribution of data is a GMM technically superior for market segmentation?
- **Trade-Off Management:** How do you navigate the trade-offs between model mathematical complexity, execution latency, infrastructure hosting cost, and regulatory explainability when delivering a machine learning solution to a bank?

End of Document • Prepared for Interview Preparation at LatentView Analytics.